EM Algorithm for Gaussian Mixture Models

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Assume $X_i \sim \sum_{k=1}^{K} \pi_k \, N(\mu_k, \sigma_k^2)$, with $\sum_{k=1}^{K} \pi_k = 1$.

Introduce the latent variable $Z_i \in \{1, \ldots, K\}$, where $Z_i = k$ means that observation i belongs to component k.

Complete-data likelihood

If both X and the component memberships Z were observed, then

$$P(X, Z \mid \theta) = \prod_{i=1}^{n} P(X_i = x_i, Z_i \mid \theta).$$

For each observation, if $Z_i = k$,

$$P(X_i = x_i, Z_i = k \mid \theta) = P(Z_i = k \mid \theta) P(X_i = x_i \mid Z_i = k, \theta).$$

For a Gaussian mixture model,

$$P(Z_i = k \mid \theta) = \pi_k$$

and

$$P(X_i = x_i \mid Z_i = k, \theta) = f(x_i; \mu_k, \sigma_k^2).$$

Therefore,

$$P(X_i = x_i, Z_i = k \mid \theta) = \pi_k f(x_i; \mu_k, \sigma_k^2).$$

Because only one component can generate observation i, we can use the indicator $I(Z_i = k)$ to write the contribution of observation i as

$$\prod_{k=1}^{K} \left[\pi_k f(x_i; \mu_k, \sigma_k^2)\right]^{I(Z_i = k)}.$$

Hence, the complete-data likelihood is

$$P(X, Z \mid \theta) = \prod_{i=1}^{n} \prod_{k=1}^{K} \left[\pi_k f(x_i; \mu_k, \sigma_k^2)\right]^{I(Z_i = k)}.$$

Complete-data log-likelihood

Taking logs,

$$\begin{aligned}\log P(X, Z \mid \theta) &= \log \left\{ \prod_{i=1}^{n} \prod_{k=1}^{K} \left[\pi_k f(x_i; \mu_k, \sigma_k^2)\right]^{I(Z_i=k)} \right\} \\ &= \sum_{i=1}^{n} \sum_{k=1}^{K} I(Z_i = k) \log \left[\pi_k f(x_i; \mu_k, \sigma_k^2)\right] \\ &= \sum_{i=1}^{n} \sum_{k=1}^{K} I(Z_i = k) \left[\log \pi_k + \log f(x_i; \mu_k, \sigma_k^2)\right].\end{aligned}$$

E-step

At iteration t, we use the current parameter values $\theta^{(t)}$ and take the expectation of the complete-data log-likelihood with respect to the conditional distribution of the missing variables Z given the observed data X:

$$Q(\theta, \theta^{(t)}) = E_{Z|X,\theta^{(t)}} \left[\log P(X, Z \mid \theta)\right].$$

Substituting the complete-data log-likelihood,

$$Q(\theta, \theta^{(t)}) = E_{Z|X,\theta^{(t)}} \left[\sum_{i=1}^{n} \sum_{k=1}^{K} I(Z_i = k) \left(\log \pi_k + \log f(x_i; \mu_k, \sigma_k^2)\right) \right].$$

Using linearity of expectation,

$$Q(\theta, \theta^{(t)}) = \sum_{i=1}^{n} \sum_{k=1}^{K} E_{Z|X,\theta^{(t)}} \left[I(Z_i = k)\right] \left[\log \pi_k + \log f(x_i; \mu_k, \sigma_k^2)\right].$$

The only random quantity here with respect to $Z \mid X$ is the indicator $I(Z_i = k)$. Its conditional expectation is

$$E_{Z|X,\theta^{(t)}} \left[I(Z_i = k)\right] = P(Z_i = k \mid X_i = x_i, \theta^{(t)}).$$

To compute this posterior probability, we use Bayes' rule:

$$P(Z_i = k \mid X_i = x_i, \theta^{(t)}) = \frac{P(X_i = x_i \mid Z_i = k, \theta^{(t)}) P(Z_i = k \mid \theta^{(t)})}{P(X_i = x_i \mid \theta^{(t)})}.$$

For the numerator,

$$P(Z_i = k \mid \theta^{(t)}) = \pi_k^{(t)}$$

and

$$P(X_i = x_i \mid Z_i = k, \theta^{(t)}) = f\left(x_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)}\right).$$

For the denominator, we marginalise over all possible components:

$$P(X_i = x_i \mid \theta^{(t)}) = \sum_{j=1}^{K} P(X_i = x_i \mid Z_i = j, \theta^{(t)}) P(Z_i = j \mid \theta^{(t)}).$$

Therefore,

$$P(X_i = x_i \mid \theta^{(t)}) = \sum_{j=1}^{K} \pi_j^{(t)} f\left(x_i; \mu_j^{(t)}, (\sigma_j^2)^{(t)}\right).$$

Substituting into Bayes' rule,

$$P(Z_i = k \mid X_i = x_i, \theta^{(t)}) = \frac{\pi_k^{(t)} f\left(x_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)}\right)}{\sum_{j=1}^{K} \pi_j^{(t)} f\left(x_i; \mu_j^{(t)}, (\sigma_j^2)^{(t)}\right)}.$$

Therefore,

$$Q(\theta, \theta^{(t)}) = \sum_{i=1}^{n} \sum_{k=1}^{K} P(Z_i = k \mid X_i = x_i, \theta^{(t)}) \left[\log \pi_k + \log f(x_i; \mu_k, \sigma_k^2)\right].$$

The key idea in the E-step is therefore

$$\boxed{I(Z_i = k) \xrightarrow{E_{Z|X,\theta^{(t)}}} P(Z_i = k \mid X_i = x_i, \theta^{(t)})}$$

M-step

Maximise $Q(\theta, \theta^{(t)})$ with respect to the parameters. The resulting updates are

$$\boxed{\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)})}$$

$$\boxed{\mu_k^{(t+1)} = \frac{\sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)}) x_i}{\sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)})}}$$

and, if the variances are unknown,

$$\boxed{(\sigma_k^2)^{(t+1)} = \frac{\sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)}) \left(x_i - \mu_k^{(t+1)}\right)^2}{\sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)})}}$$

Then repeat the E-step and M-step until convergence.

Appendix: M-step calculations

For compactness, define

$$r_{ik}^{(t)} = P(Z_i = k \mid X_i = x_i, \theta^{(t)}).$$

Then the Q-function can be written as

$$Q(\theta, \theta^{(t)}) = \sum_{i=1}^{n} \sum_{k=1}^{K} r_{ik}^{(t)} \left[\log \pi_k + \log f(x_i; \mu_k, \sigma_k^2) \right].$$

For a Gaussian density,

$$\log f(x_i; \mu_k, \sigma_k^2) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma_k^2) - \frac{(x_i - \mu_k)^2}{2\sigma_k^2}.$$

Hence,

$$Q(\theta, \theta^{(t)}) = \sum_{i=1}^{n} \sum_{k=1}^{K} r_{ik}^{(t)} \left[\log \pi_k - \frac{1}{2} \log(2\pi) - \frac{1}{2} \log(\sigma_k^2) - \frac{(x_i - \mu_k)^2}{2\sigma_k^2} \right].$$

Update for the mixture probabilities π_k

We maximise the part of Q that depends on π_k:

$$Q_\pi = \sum_{i=1}^{n} \sum_{k=1}^{K} r_{ik}^{(t)} \log \pi_k,$$

subject to

$$\sum_{k=1}^{K} \pi_k = 1.$$

Introduce the Lagrange multiplier λ:

$$\mathcal{L} = \sum_{i=1}^{n} \sum_{k=1}^{K} r_{ik}^{(t)} \log \pi_k + \lambda \left(\sum_{k=1}^{K} \pi_k - 1 \right).$$

Differentiate with respect to π_k:

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{\sum_{i=1}^{n} r_{ik}^{(t)}}{\pi_k} + \lambda.$$

Set equal to zero:

$$\frac{\sum_{i=1}^{n} r_{ik}^{(t)}}{\pi_k} + \lambda = 0.$$

Therefore,

$$\pi_k = -\frac{1}{\lambda} \sum_{i=1}^{n} r_{ik}^{(t)}.$$

Now sum over k:

$$\sum_{k=1}^{K} \pi_k = -\frac{1}{\lambda} \sum_{k=1}^{K} \sum_{i=1}^{n} r_{ik}^{(t)}.$$

Since

$$\sum_{k=1}^{K} r_{ik}^{(t)} = 1$$

for every observation i,

$$1 = -\frac{1}{\lambda} \sum_{i=1}^{n} 1 = -\frac{n}{\lambda}.$$

Hence,

$$\lambda = -n.$$

Substituting this back gives

$$\boxed{\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^{n} r_{ik}^{(t)}}$$

or, equivalently,

$$\boxed{\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)})}.$$

Update for the means μ_k

For a fixed component k, keep only the terms in Q that depend on μ_k:

$$Q_{\mu_k} = -\frac{1}{2\sigma_k^2} \sum_{i=1}^{n} r_{ik}^{(t)} (x_i - \mu_k)^2.$$

Differentiate with respect to μ_k:

$$\frac{\partial Q_{\mu_k}}{\partial \mu_k} = -\frac{1}{2\sigma_k^2} \sum_{i=1}^{n} r_{ik}^{(t)} \frac{\partial}{\partial \mu_k} (x_i - \mu_k)^2.$$

Since

$$\frac{\partial}{\partial \mu_k} (x_i - \mu_k)^2 = -2(x_i - \mu_k),$$

we obtain

$$\frac{\partial Q_{\mu_k}}{\partial \mu_k} = \frac{1}{\sigma_k^2} \sum_{i=1}^{n} r_{ik}^{(t)} (x_i - \mu_k).$$

Set equal to zero:

$$\sum_{i=1}^{n} r_{ik}^{(t)}(x_i - \mu_k) = 0.$$

Expanding,

$$\sum_{i=1}^{n} r_{ik}^{(t)} x_i - \mu_k \sum_{i=1}^{n} r_{ik}^{(t)} = 0.$$

Therefore,

$$\boxed{\mu_k^{(t+1)} = \frac{\sum_{i=1}^{n} r_{ik}^{(t)} x_i}{\sum_{i=1}^{n} r_{ik}^{(t)}}}$$

or

$$\boxed{\mu_k^{(t+1)} = \frac{\sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)}) x_i}{\sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)})}}.$$

Update for the variances σ_k^2

For a fixed component k, keep only the terms involving σ_k^2:

$$Q_{\sigma_k^2} = -\frac{1}{2} \sum_{i=1}^{n} r_{ik}^{(t)} \left[\log(\sigma_k^2) + \frac{(x_i - \mu_k)^2}{\sigma_k^2} \right].$$

Differentiate with respect to σ_k^2:

$$\frac{\partial Q_{\sigma_k^2}}{\partial \sigma_k^2} = -\frac{1}{2} \sum_{i=1}^{n} r_{ik}^{(t)} \left[\frac{1}{\sigma_k^2} - \frac{(x_i - \mu_k)^2}{(\sigma_k^2)^2} \right].$$

Set equal to zero:

$$\sum_{i=1}^{n} r_{ik}^{(t)} \left[\frac{1}{\sigma_k^2} - \frac{(x_i - \mu_k)^2}{(\sigma_k^2)^2} \right] = 0.$$

Multiply through by $(\sigma_k^2)^2$:

$$\sigma_k^2 \sum_{i=1}^{n} r_{ik}^{(t)} - \sum_{i=1}^{n} r_{ik}^{(t)} (x_i - \mu_k)^2 = 0.$$

Therefore,

$$\sigma_k^2 = \frac{\sum_{i=1}^{n} r_{ik}^{(t)} (x_i - \mu_k)^2}{\sum_{i=1}^{n} r_{ik}^{(t)}}.$$

Using the newly updated mean $\mu_k^{(t+1)}$,

$$\boxed{(\sigma_k^2)^{(t+1)} = \frac{\sum_{i=1}^{n} r_{ik}^{(t)} \left(x_i - \mu_k^{(t+1)}\right)^2}{\sum_{i=1}^{n} r_{ik}^{(t)}}}$$

or, equivalently,

$$\boxed{(\sigma_k^2)^{(t+1)} = \frac{\sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)}) \left(x_i - \mu_k^{(t+1)}\right)^2}{\sum_{i=1}^{n} P(Z_i = k \mid X_i = x_i, \theta^{(t)})}}.$$